PROBABILITY THEORY
MTM2592-Gr:2

SECOND WEEK

Assc. Prof. Dr. Ulku (BABUSCU) YESIL

ubabuscu@yildiz.edu.tr

ulkubabuscu@gmail.com

https://avesis.yildiz.edu.tr/ubabuscu

Probability Concepts

EXPERIMENT:

i) **Deterministic experiment.**

ii) **Random experiment.**

SAMPLE POINT & SAMPLE SPACE:

$$S = \{H, T\}$$

Sample space

Sample point

$$S = \{(x, y) \mid x = 1, 2, .., 6; y = 1, 2, .., 6\}$$
$$= \{(1,1), (1, 2), ..., (6, 6)\}$$

Sample point

DISCRETE AND CONTINUOUS SAMPLE SPACE:

$$S = \{1, 2, 3, 4, 5, 6\} \longrightarrow \text{discrete sample space}$$

$$S = \{x \mid 2 < x < 5\} \longrightarrow \text{continuous sample space}$$

EVENT:

An event is a subset of a sample space. A sample point or several sample points are called an event.

If an event corresponds to a single point of a sample space, it is said to be a simple (elementary) event.

If it corresponds to more than one point, then it is called a compound event.

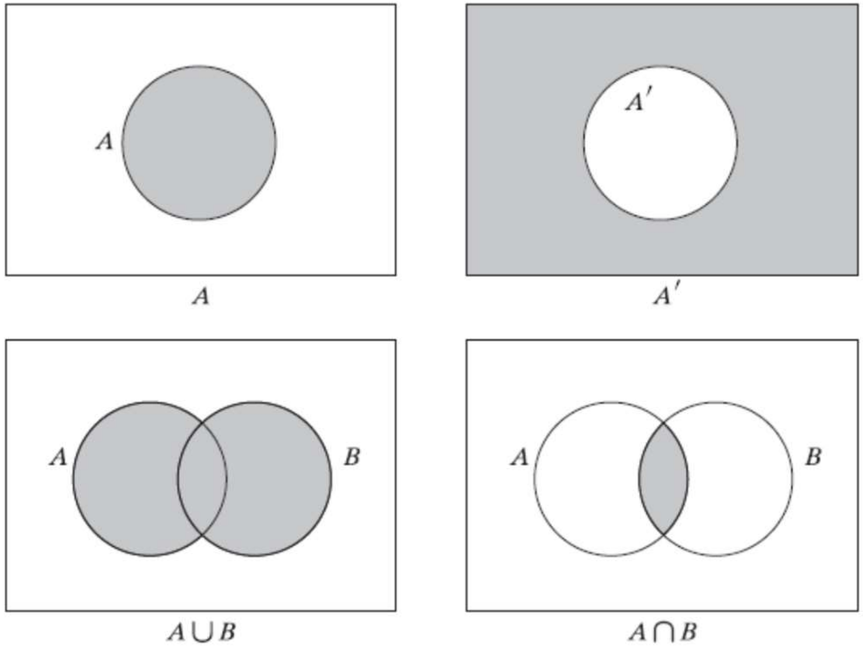


Figure 3. Venn diagrams.

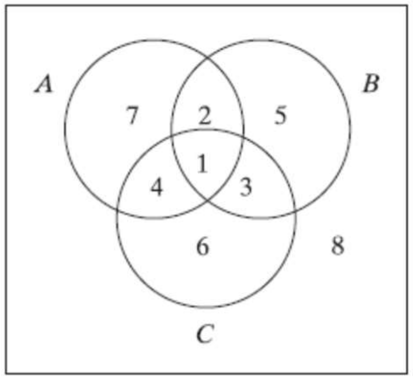


Figure 4. Venn diagram.

DEFINITION MUTUALLY EXCLUSIVE EVENTS. *Two events having no elements in common are said to be **mutually exclusive**.*

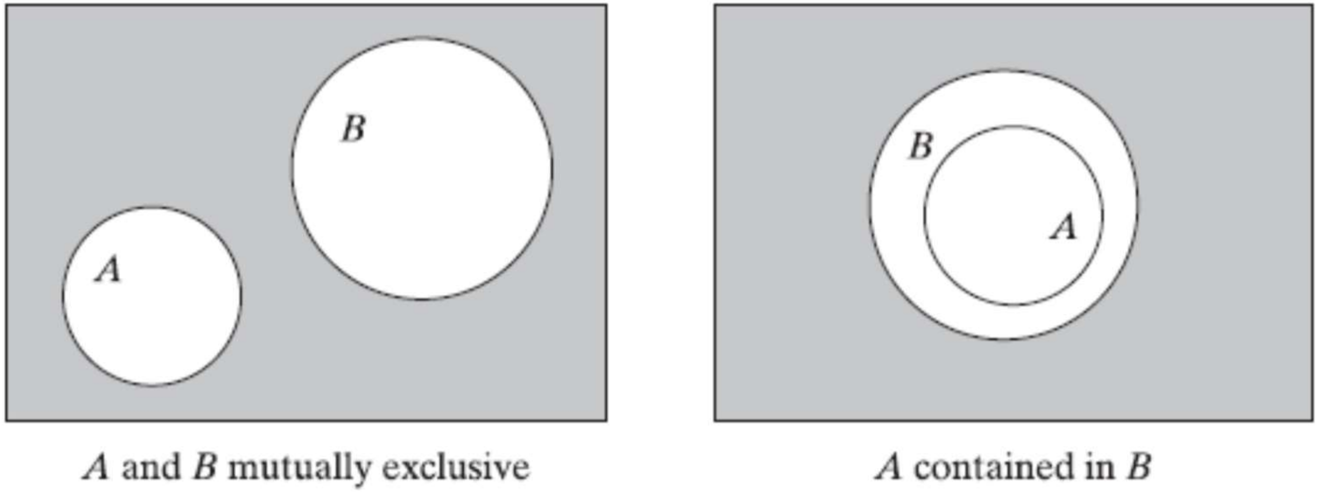


Figure 5. Diagrams showing special relationships among events.

Example: If a random experiment consist of tossing a balanced die, then the sample space is

$$S = \{1, 2, 3, 4, 5, 6\}$$

$A_1 = \{1, 3, 5\}$ ➡ The event of getting odd numbers

$A_2 = \{2, 4, 6\}$ ➡ The event of getting even numbers

$A_3 = \{1, 2\}$ ➡ The event of getting a number smaller than 3

Example: Lifetime of produced bulbs

$$S = \{x \mid 0 \le x < \infty\}$$

$B_1 = \{x \mid x > 200\}$ ➡ The event of lasting longer than 200 hours

Example: Let a balanced coin be tossed three times. Construct a sample space and an event that the coin shows head at least two times.

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$A = \{HHH, HHT, HTH, THH\}$$

Example: Let $S = \{x | 1 \le x \le 5\}$ be a sample space and

$A = \{x | 1 \le x \le 4\}$, $B = \{x | 2 < x \le 5\}$ be two events. Construct the events

$$A \cup B = \{x | 1 \le x \le 5\}$$

$$A \cup \overline{B} = \{x | 1 \le x \le 4\}$$

$$\overline{A} \cap B = \{x | 4 < x \le 5\}$$

$$\overline{A \cap B} = \{x | 1 \le x \le 2 \text{ or } 4 < x \le 5\}$$

THE DEFINITIONS OF THE PROBABILITY:

Conventional Definition: Let S be a sample space that contains equally likely outcomes and A be an event defined on S. The probability of the event A is

$$P(A) = \frac{n(A)}{n(S)}$$

Ex: What is the probability that a balanced die comes up greater than 4?

$$S = \{1, 2, 3, 4, 5, 6\} \quad A = \{5, 6\} \quad P(A) = \frac{n(A)}{n(S)} = \frac{2}{6}$$

Frequency Definition: If N is the number of trials of an experiment and n is the number of times the event A happens, then

$$p(A) = \lim_{N \to \infty} \frac{n}{N}$$

Subjective Definition:

x: The chance of occurrence of an event A
y: The chance of non-occurrence of an event A

$$\frac{p(A)}{1 - p(A)} = \frac{x}{y} \Rightarrow p(A) = \frac{x}{x + y} \qquad \text{(subjective chance)}$$

Ex: An expert considers that the chance of Besiktas's winning the cup is 2:5. According to this expert what is the probability that Besiktas will win the cup?

P=2/(2+5)=2/7

PROBABILITY AXIOMS:

Definition: A real valued function P, defined on a sample space S is called a probability function if the following axioms are satisfied.

1) $\boxed{0 \le P(A) \le 1}$ for all events A

2) P(S)=1

3) If A_1, A_2,...., A_n are disjoint events (mutually exclusive), then

$$\boxed{P(A_1 \cup A_2 \cup A_3 \ldots.) = P(A_1) + P(A_2) + P(A_3) + \ldots\ldots}$$

For two disjoint events ⇒ $\boxed{P(A \cup B) = P(A) + P(B)}$

Theorem 1: $\boxed{P(\varnothing) = 0}$ where $\varnothing$ is an empty set.

Proof: A is any set.

$$A \cap \varnothing = \varnothing \quad \text{and} \quad A \cup \varnothing = A$$

$$\boxed{P(A) = P(A \cup \varnothing) = P(A) + P(\varnothing)} \Longrightarrow P(\varnothing) = 0$$

Theorem 2: $P(\overline{A}) = 1 - P(A)$ where $\overline{A}$ is the complement of A.

Proof: $S = A \cup \overline{A} \quad \text{and} \quad A \cap \overline{A} = \varnothing$

$$1 = P(S) = P(A \cup \overline{A}) = P(A) + P(\overline{A})$$

$$P(\overline{A}) = 1 - P(A)$$

$\boxed{\text{Theorem 3: If } \quad A \subset B \quad \text{then} \quad P(A) \le P(B)}$

Proof: $\left(B \setminus A\right) \cap A = \varnothing, \quad \left(B \setminus A\right) \cup A = B$

$$P\left(\left(B \setminus A\right) \cup A\right) = P\left(B \setminus A\right) + P(A) = P(B)$$

$$P(B/A) \ge 0 \implies P(B) \ge P(A)$$

Remark: $A \subset S \quad \text{then} \quad P(A) \le P(S)$

$$P(S) = 1, \; P(A) \ge 0$$

$$\text{then} \quad 0 \le P(A) \le P(S) \implies 0 \le P(A) \le 1$$

Theorem 4: $\boxed{P(A/B) = P(A) - P(A \cap B)}$

Proof: $A = (A/B) \cup (A \cap B)$ since $(A/B) \cap (A \cap B) = \varnothing$

$$P(A) = P\big((A/B) \cup (A \cap B)\big) = P(A/B) + P(A \cap B)$$

$$P(A/B) = P(A) - P(A \cap B)$$

Theorem 5: If A and B are any event then

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

Proof: $A \cup B = (A/B) \cup B$

By using Theorem 4

$$P(A \cup B) = P(A/B) + P(B) = P(A) - P(A \cap B) + P(B)$$

$$= P(A) + P(B) - P(A \cap B)$$

THEOREM If A, B, and C are any three events in a sample space S, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C)$$
$$- P(B \cap C) + P(A \cap B \cap C)$$

Proof

$$\begin{aligned} P(A \cup B \cup C) &= P[A \cup (B \cup C)] \\ &= P(A) + P(B \cup C) - P[A \cap (B \cup C)] \\ &= P(A) + P(B) + P(C) - P(B \cap C) \\ &\quad - P[A \cap (B \cup C)] \end{aligned}$$

$$\Downarrow$$

$$\begin{aligned} P[A \cap (B \cup C)] &= P[(A \cap B) \cup (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P[(A \cap B) \cap (A \cap C)] \\ &= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \end{aligned}$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C)$$
$$- P(B \cap C) + P(A \cap B \cap C)$$

Theorem 6: If $A_1, A_2, \ldots A_n$ are any event then;

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = \sum_{i=1}^{n} P(A_i) - \sum_{1 \le i < j \le n} P(A_i \cap A_j) + \sum_{1 \le i < j < k \le n} P(A_i \cap A_j \cap A_k) - \ldots\ldots + (-1)^{n-1} P(A_1 \cap A_2 \ldots \cap A_n)$$

BOOLE INEQUALITY: ➡ $P(A_1 \cup A_2) \le P(A_1) + P(A_2)$

If $A_1, A_2, \ldots A_n$ are any event ➡ $P\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} P(A_i)$

BONFERONNI INEQUALITY ➡ $P(A_1 \cap A_2) \ge P(A_1) + P(A_2) - 1$

General form of this inequality ➡ $P\left(\bigcap_{i=1}^{n} A_i\right) \ge \sum_{i=1}^{n} P(A_i) - 1$

Bonferroni's inequality is used to obtain a lower limit where it is impossible to calculate the probability of intersection.

Example: If P(A)=0.90, P(B)=0.95 create a lower limit for $P(A \cap B)$

Solution: $P(A \cap B) \geq P(A) + P(\mathrm{B}) - 1 = 0.90 + 0.95 - 1 = 0.85$

Theorem 7: If A and B are any events, than

$$\boxed{\mathrm{P(A) = P(A \cap B) + P(A \cap \overline{B})}}$$

Example: A card is randomly selected from a deck of 52 cards. Let;

$$\left.\begin{aligned} A &= \{The\ \ card\ \ is\ \ Spades\} \\ B &= \{The\ \ card\ \ is\ \ colorful(Jack,\ \ Queen\ \ or\ \ King)\} \end{aligned}\right\}$$

Find P(A), P(B) and $P(A \cap B)$

Note:

Jack $\rightarrow$ Bacak	Hearts $\rightarrow$ Kupa
Queen $\rightarrow$ Kız	Spades $\rightarrow$ Maça
King $\rightarrow$ Papaz	Diamonds $\rightarrow$ Karo
Ace $\rightarrow$ As	Clubs $\rightarrow$ Sinek

Solution:

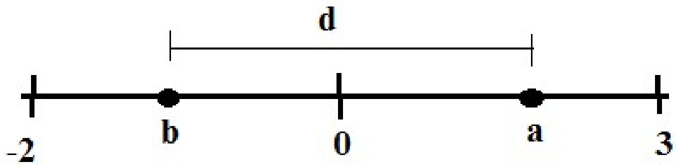
Example: Two items are randomly chosen from a set of 12 items of which 4 are defective.

$$\left.\begin{aligned} A &= \{Both\ of\ the\ chosen\ items\ are\ defective\} \\ B &= \{Neither\ of\ the\ chosen\ items\ are\ defective\} \\ C &= \{At\ least\ one\ of\ the\ chosen\ items\ are\ defective\} \end{aligned}\right\}$$

Find P(A), P(B) and P(C).

Solution:

Example: Points a and b are randomly selected on a R real line as shown below where $-2 \leq b \leq 0 \quad and \quad 0 \leq a \leq 3$. What is the probability that the distance d between a and b is longer than 3?



Solution:

Example: The probability that a student passes from mathematics is 4/5, the probability that a student passes from physics is 2/3 and the probability that the student passes from both subjects is ½. Find the probability that the student

a) Fails physics,
b) Passes mathematics or physics,
c) Passes mathematics or fails physics.

Solution:

Example: If two dies are rolled, what is the probability that the total number comes up 9?

Solution:

Example: A coin is tossed three times.

$$A = \left\{ Comes\ up\ head\ at\ least\ two\ times \right\}$$

$$B = \left\{ Comes\ up\ head\ at\ the\ \sec ond\ tos\sin g \right\}$$

$$C = \left\{ Comes\ up\ head\ only\ at\ the\ \sec ond\ tos\sin g \right\}$$

a) Are A and B disjoint events,
b) Are A and C disjoint events,
c) Are B and C disjoint events,
d) Find $P(A \cup B),\ \ P(A \cup C),\ \ P(B \cup C)$

Solution:

CONDITIONAL PROBABILITY:

Let A and B are two events defined on a sample space S. Then, the probability of the event A occuring when it is known that the event B has occured is called a conditional probability and it is denoted by

$$\boxed{P(A|B)}$$

We can consider

$$\boxed{P(A) = P(A|S)}$$

Thus, B is said to be a reduced sample space for this situation.

Definition: The conditional probability of A, given B denoted by $P(A|B)$ is defined by

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0}$$

Example: The people in a small town are categorized as shown in the table:

	Employed (E)	Unemployed (U)
Male (M)	300	400
Female (F)	250	250

Find $P(U|M),\ P(M|U) = ?$

Example: It is known that the outcomes is an even number when a balanced die is tossed. Accordingly, what is the probability that this number is 4?

Example: 100 people are surveyed in order to determine relationship between high tension and obesity. The results are given as:

	Thin	Normal	Fat	Total
High Tension	8	10	22	40
Non-high Tension	24	16	20	60
Total	32	26	42	100

Find the probability that a randomly selected person

a) is fat,
b) has high tension,
c) is fat given that he/she has high tension
d) has high tension given that he/she is fat.

Example: Let two dies be rolled. What is the probability that both of them come up 4 given that they comes up 8 in total.
Solution:

Example: Two balls successively selected without replacing them back, from a bag that contains 3 green balls and 5 red balls. What is the probability that the second chosen ball is green given that the first ball is green?
Solution:

MULTIPLICATION RULE FOR CONDITIONAL PROBABILITY:

* The probability of occurrence of the intersection of two events can be calculated using the product rule.

$$P(A_1|A_2)=\frac{P(A_1\cap A_2)}{P(A_2)} \Rightarrow P(A_1\cap A_2)=P(A_1|A_2).P(A_2)$$

$$P(A_2)>0$$

or

$$P(A_2|A_1)=\frac{P(A_1\cap A_2)}{P(A_1)} \Rightarrow P(A_1\cap A_2)=P(A_2|A_1).P(A_1)$$

$$P(A_1)>0$$

*In the case of three events;

$$P(A_3|A_1\cap A_2)=\frac{P(A_1\cap A_2\cap A_3)}{P(A_1\cap A_2)}$$

$$P(A_1\cap A_2\cap A_3)=P(A_1\cap A_2)P(A_3|A_1\cap A_2)$$

$$P(A_1\cap A_2\cap A_3)=P(A_1)P(A_2|A_1)P(A_3|A_1\cap A_2)$$

*This formula can be generalized for n events $A_1, A_2, \dots A_n$,

$$P(A_1\cap A_2\cap\ldots\cap A_n)=P(A_1).P(A_2|A_1).P(A_3|A_1\cap A_2)\ldots.P(A_n|A_1\cap A_2\cap\ldots A_{n-1})$$

$$P\left(\bigcap_{i=1}^{n}A_i\right)=P(A_1).P(A_2|A_1).P(A_3|A_1\cap A_2)\ldots.P\left(A_n\left|\left(\bigcap_{i=1}^{n-1}A_i\right)\right.\right)$$

Example: 4 bulbs are defective in a box of 18 bulbs. 2 bulbs are successively selected without replacing them back. What is the probability that

a) Both bulbs are defective,
b) Both bulbs are non-defective,
c) The first chosen bulb is defective and the second one is non-defective,
d) The first chosen bulb is non-defective and the other one is defective,

Solution:

Example: 4 bulbs are defective in a box of 15 bulbs. 3 bulbs are successively selected without replacing them back. What is the probability that all bulbs are non-defective?

Solution:

Example: A box contains 8 blue balls and 6 red balls. 3 balls are selected successively from the box. Find the probability that the first ball is red, the second ball is blue and third one is red, if 3 balls are chosen

a) Without replacing them back,

b) Replacing them back.

Solution:

DISCRETIZATION OF SAMPLE SPACE:

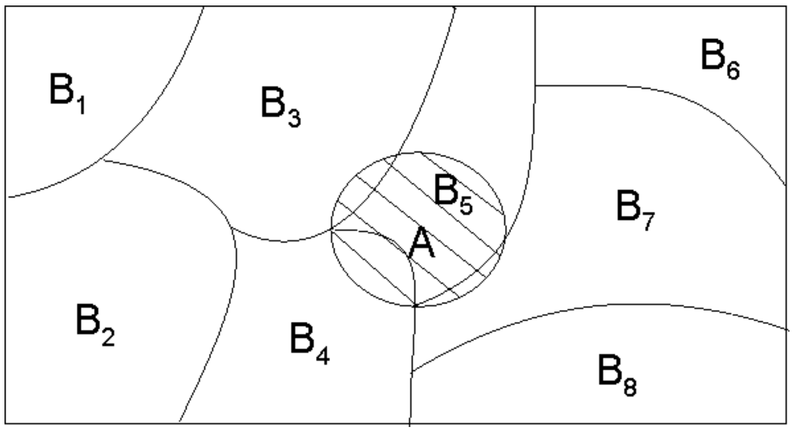
Let $B_i \subset S,$

a) $P(B_i \cap B_j) = \varnothing \quad i \neq j$

b) $\bigcup_{i=1}^{n} B_i = S$

c) $P(B_i) > 0 \quad \text{for all } i$

then B_1, B_2,… B_n events show a discretization of the sample space S. The figure shows discretization of S for n = 8.



Theorem: According to the definition given above,

$$\boxed{P(B_1) + P(B_2) + \ldots + P(B_n) = 1}$$

Theorem: If B_i (i=1,2,…,n) is a disretization of a sample space S, then

$$\boxed{P(A) = \sum_{i=1}^{n} P(B_i \cap A) = \sum_{i=1}^{n} P(B_i).P(A|B_i)}$$

for any events A in S.

Proof:

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \ldots \cup (A \cap B_n)$$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots + P(A \cap B_n)$$

Since $A \cap B_i$ are disjoint events

$$P(A \cap B_i) = P(B_i)P(A|B_i)$$

$$P(A) = P(B_1)P(A|B_1) + \ldots\ldots + P(B_n)P(A|B_n)$$

Example: A type of machine is produced in the factories A, B and C. The number of machines produced in A is 2 times of machines produced in B and also in C. It is known that 2% of machines produced in A as well as in B, and 4% of machines produced in C are defective. If a machine is randomly selected from the store in which all produced machines put, what is the probability that the machine is defective?

Solution:

INDEPENDENT EVENTS:

Definition: The events A and B are independent if and only if

$$P(B|A) = P(B) \text{ or } P(A|B) = P(A)$$

Otherwise, A and B are dependent.

Theorem: The events A and B are independent if and only if

$$\boxed{P(A \cap B) = P(A).P(B)}$$

Theorem: If A and B are independent events that $(P(A) \neq 0 \text{ and } P(B) \neq 0)$ then A and B have at least one common sample point. So, $A \cap B \neq \varnothing$

Theorem: If A is independent of B, then B is independent of A.

$$P(A|B) = P(A) \iff P(B|A) = P(B)$$

Proof: $$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(A|B).P(B) = P(B|A).P(A)$$

$$P(A|B) = P(A), \;\; P(A) \neq 0$$

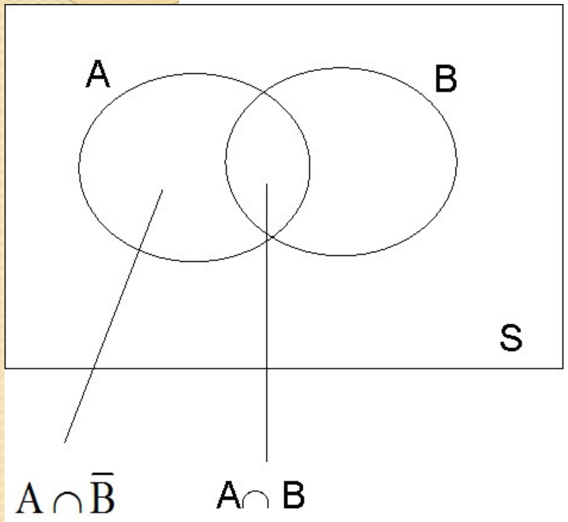
$$P(B) = P(B|A)$$

Theorem: If A and B are independent, then A and $\overline{B}$, $\overline{A}$ and B; $\overline{A}$ and $\overline{B}$ are also independent.

Proof: If A and B are independent, then A and $\bar{B}$ are also independent

$$A, B \subset S$$

$$P(A|B) = P(A) \Rightarrow P(A|\bar{B}) = P(A)$$



$$A = (A \cap B) \cup (A \cap \bar{B})$$

$$(A \cap B) \cap (A \cap \bar{B}) = \varnothing$$

$$P((A \cap B) \cup (A \cap \bar{B})) = P(A \cap B) + P(A \cap \bar{B})$$

$$P(A) = P(A)P(B) + P(A \cap \bar{B})$$

$$P(A)[1 - P(B)] = P(A \cap \bar{B})$$

$$P(A)\,P(\bar{B}) = P(A \cap \bar{B})$$

And hence A and $\bar{B}$ are independent.

TOTAL INDEPENDENCY

Definition: Events $A_1, A_2, \ldots, A_n$ are total independent if and only if any 2,3,…,(n-1) of these events are independent and

$$\boxed{P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i)}$$

For example, if n=3,

$$\left.\begin{aligned}
P(A_1 \cap A_2 \cap A_3) &= P(A_1).P(A_2).P(A_3) \\
P(A_1 \cap A_2) &= P(A_1).P(A_2) \\
P(A_1 \cap A_3) &= P(A_1).P(A_3) \\
P(A_2 \cap A_3) &= P(A_2).P(A_3)
\end{aligned}\right\} \text{for total independency}$$

Example: It is estimated that in Turkey, 22% of bachelors are woman and 15% of bachelors have economy degree. On the other hand, 7% of bachelors have economy degree and woman. Decide whether the events «be bachelor woman» and «be bachelor having economy degree» are independent or not.

Solution:

Example: Figure 8 shows a Venn diagram with probabilities assigned to its various regions. Verify that A and B are independent, A and C are independent, and B and C are independent, but A, B, and C are not independent.

Solution

As can be seen from the diagram, $P(A) = P(B) = P(C) = \frac{1}{2}$, $P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4}$, and $P(A \cap B \cap C) = \frac{1}{4}$. Thus,

$$P(A) \cdot P(B) = \frac{1}{4} = P(A \cap B)$$

$$P(A) \cdot P(C) = \frac{1}{4} = P(A \cap C)$$

$$P(B) \cdot P(C) = \frac{1}{4} = P(B \cap C)$$

but

$$P(A) \cdot P(B) \cdot P(C) = \frac{1}{8} \neq P(A \cap B \cap C)$$

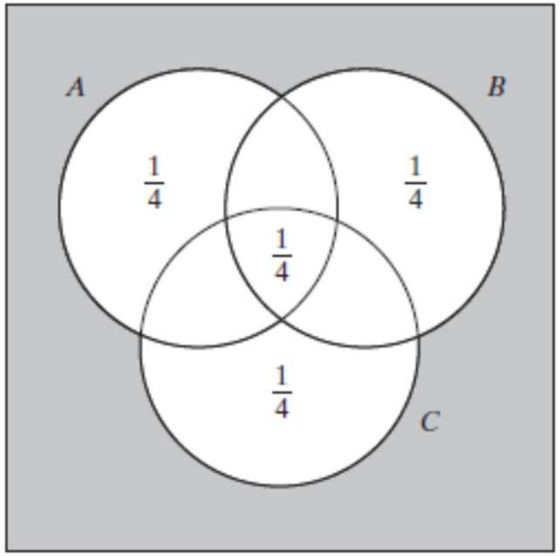


Figure 8. Venn diagram

Example: Let a coin be tossed two times.

$$A = \{The\ coin\ shows\ tail\ at\ the\ first\ tossing\}$$

$$B = \{The\ coin\ shows\ tail\ at\ the\ second\ tossing\}$$

$$C = \{The\ coin\ shows\ tail\ only\ one\ times\}$$

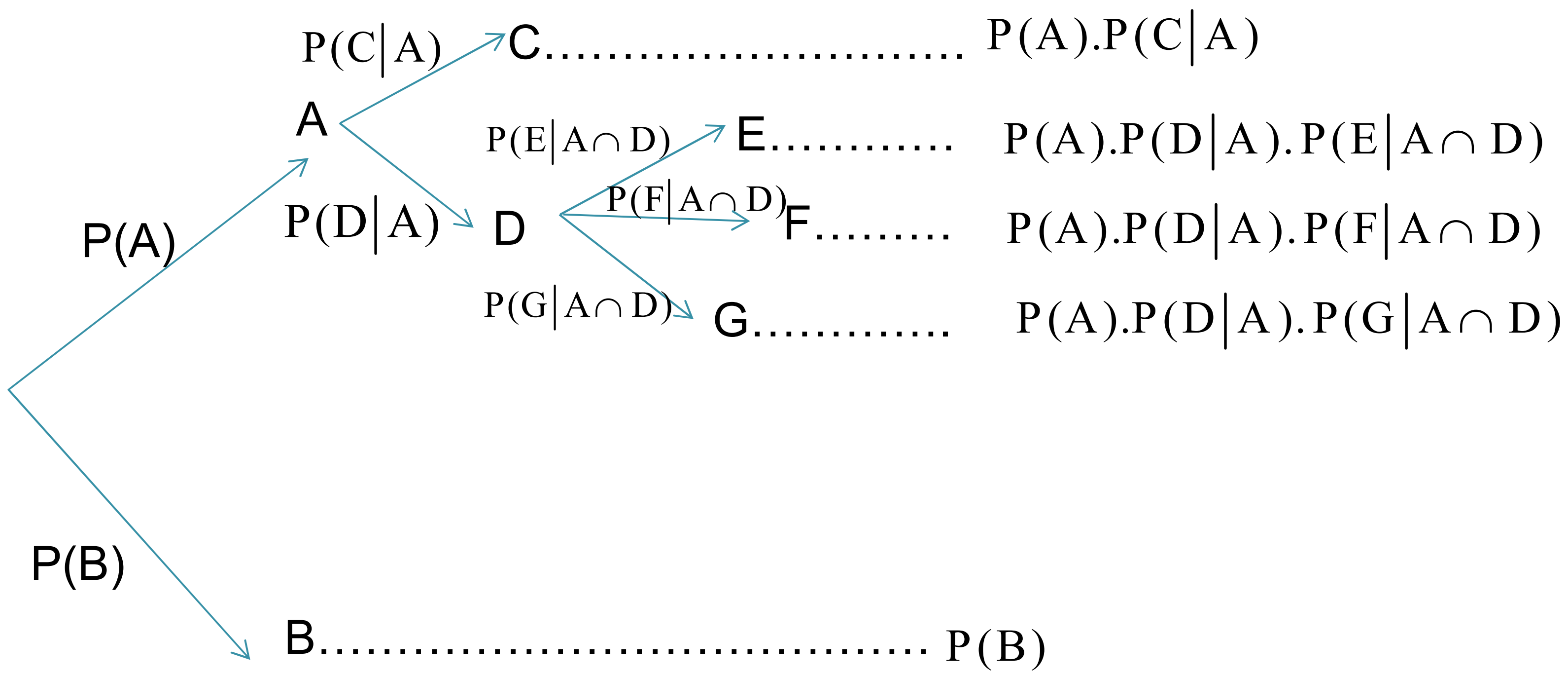
Examine the total independency.

Solution:

Example: The probability of hitting a target is 95% for the person A and 82% for the person B. What is the probability that the target will be hit if both person try to hit the same target?

Solution:

DETERMINATION OF CONDITIONAL PROBABILITY BY TREE DIAGRAMS:



Example: There are 3 white balls and 5 red balls in a box. 2 balls are successively selected from the box, without replacing them back. Find the probability that two balls are different coloured by using tree diagram.

Solution:

Example: There are 4 black and 6 red balls in the first box and 3 black and 2 red balls in the second box. If we randomly select a box, then randomly select a ball from the chosen box, what is the probability that

a) The chosen ball is black,
b) The chosen box is the first one given that a black ball is selected?

Solution:

BAYES' THEOREM:

Let B_1, B_2,… B_n be discretization of a sample space S and A be an event in S with $P(A) \neq 0$
Then,

$$\boxed{P(B_r|A) = \frac{P(B_r).P(A|B_r)}{\sum_{i=1}^{n} P(B_i)P(A|B_i)}}$$

Proof: Note:

$$\boxed{P(A) = \sum_{i=1}^{n} P(B_i \cap A) = \sum_{i=1}^{n} P(B_i).P(A|B_i)}$$

$$P(B_r|A) = \frac{P(A \cap B_r)}{P(A)} = \frac{P(B_r).P(A|B_r)}{\sum_{i=1}^{n} P(B_i)P(A|B_i)}$$

Note: Bayes' Theorem is based on obtaining an unknown $P(B|A)$ conditional probability using the known $P(A|B)$ conditional probability.

Example: In a factory, three machines B_1, B_2, B_3 , make 30%, 45% and 25%, respectively of the products. It is known that 2%, 3% and 2% of the products made by each machine, respectively are defective.

a) What is the probability that a randomly selected product is defective?
b) If a randomly selected product is defective, what is the probability that it was made by machine B_3?

Solution:

Example: Members of a consulting firm rent cars from three companies, 60% from Company 1, 30% from Company 2, and 10% from Company 3. If 9% of vehicles from Company 1, 20% of vehicles from Company 2 and 6% of vehicles from Company 3 require maintenance,

a) What is the probability that a car rented to the company will require maintenance?

b) If the car rented to the company requires maintenance, what is the probability that this vehicle came from Company 2?

Solution: